TEMS Academy of Olympiad Math
Functional Equations Problem Set

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Problem 1. (Korea Summer Program Junior P2) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(y^2 f(x)) = xyf(y)$$

for all $x, y \in \mathbb{R}$.

Problem 2. (Kosovo 2024 P3) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x + yf(x) + f(y)) = xy + f(f(x))$$

for all $x, y \in \mathbb{R}$.

Problem 3. (BdMO 2024 Higher Secondary P7) Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that

$$f\left(\left\lceil \frac{f(m)}{n} \right\rceil\right) = \left\lceil \frac{m}{f(n)} \right\rceil$$

for all $m, n \in \mathbb{N}$.

Problem 4. (IFEO Shortlist 2020 A1) Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$1 + f^{f(a)+b}(a) = f(a) + f(a + b)$$

for all $a, b \in \mathbb{R}$.

Problem 5. (USAJMO 2024 P5) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x^2 - y) + 2yf(x) = f(f(x)) + f(y)$$

for all $x, y \in \mathbb{R}$.

Problem 6. (FEOO Shortlist A1) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$2xyf(xy) \geq (xy)^2 + xf(x)f(y^2)$$

for all $x, y \in \mathbb{R}$.

Problem 7. (IFEO 2020 P5) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x^2 + f(x) + y) = xf(x) + f(x + y)$$

for all $x, y \in \mathbb{R}$.

Problem 8. (FEcreator handout) Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f(2x + 2f(y)) = x + f(x) + 2y$$

for all $x, y \in \mathbb{R}^+$.